

Predicting LLM Pretraining Loss Curves from Learning-rate Schedules

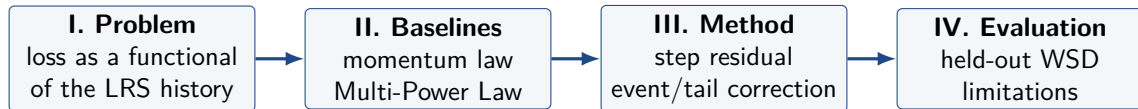
Baseline-calibrated residual modeling for cross-schedule prediction

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Code repository: <https://github.com/zyyyyyyy123/tdlt>



Core thesis

We formulate loss-curve prediction as a baseline-plus-residual transfer problem. Scaling-law baselines explain the global trajectory, while the remaining error contains a stable step-aligned component and a smaller event/tail component.

Chapter I

Problem and Protocol

What exactly is a loss-curve prediction problem, and how do we test transfer without using WSD for model selection?

Problem: Learning-rate Schedule as a Functional Input

Classical scaling laws often summarize a run by a final loss [1, 2, 6]:

$$L_{\text{final}} \approx aN^{-\alpha} + bD^{-\beta} + c,$$

where N is model size and D is the number of training tokens. Task 2 asks for a harder object: the whole trajectory under a schedule history [3, 4, 5],

$$L_t = L_{\Theta}(\eta_{\leq t}), \quad \eta_{\leq t} = (\eta_1, \eta_2, \dots, \eta_t), \quad t = 1, \dots, T.$$

Scientific question

Learning-rate schedules can have similar final or cumulative LR mass but different local decay geometry. A full-curve predictor must transfer not only final loss, but also mid-training and tail behavior.

Why This Is a Hard Transfer Problem

1. The input is history-dependent

The schedule prefix $\eta_{\leq t}$ is a sequence, not a scalar covariate. Collapsing it to cumulative LR mass, $S_1(t) = \sum_{i \leq t} \eta_i$, can erase decay timing, smoothness, and concentration.

2. Cosine-to-WSD is out-of-distribution transfer

Cosine decays smoothly across many steps, while WSD concentrates decay late in training. The test is therefore not curve interpolation, but whether a law fitted on smooth decay transfers to a different decay geometry.

3. Baselines must leave the right residual

Momentum law and Multi-Power Law provide interpretable global structure [4, 5]. The key question is whether the remaining error has transferable phase and tail geometry, rather than schedule-specific noise.

Notation and Evaluation Setting

Objects

- η_t : learning rate at optimization step t .
- $L_s(t)$: observed loss under schedule s .
- $\hat{L}_s(t)$: predicted loss under schedule s .
- T : total number of optimization steps.
- $\mathcal{T}$: evaluation step set.
- $\bar{L}_s$: mean of $L_s(t)$ over $t \in \mathcal{T}$.

Metrics

$$\text{MAE} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} |L_s(t) - \hat{L}_s(t)|$$

$$R^2 = 1 - \frac{\sum_{t \in \mathcal{T}} (L_s(t) - \hat{L}_s(t))^2}{\sum_{t \in \mathcal{T}} (L_s(t) - \bar{L}_s)^2}.$$

All reported WSD metrics use evaluation steps $t \geq 1000$, tail metrics are computed on the decay/tail window.

Course data

- Model/data setting: 100M_gpt, 20B.
- Schedules: cosine, WSD, and 8-1-1.
- Each run contains step, loss, and learning rate.
- Evaluation begins at step 1000.

Train/validation/test split

- Fit model parameters on cosine.
- Use 8-1-1 for validation and model selection.
- Keep WSD as the held-out transfer target.
- Metrics: MAE, RMSE, R^2 , tail MAE, endpoint error.

This split matches Task 2: fit on cosine LRS and evaluate prediction performance on WSD LRS.

Data Cleaning: Dense Step Grid and Forward Fill

Cleaning pipeline

- Convert each raw run to required columns: step, loss, and learning rate.
- Reindex every schedule to the dense grid $0, \dots, 33907$, giving 33908 points per run.
- Check duplicated steps and key-column NaNs before regenerating the cleaned pkl/csv files.
- Use forward fill only for isolated missing steps: copied loss and LR from the previous step.

schedule	missing step	copied from	fields
wsd_rope	20815	20814	loss, LR
cosine_rope	22493	22492	loss, LR
811_rope	none	—	—

Why ffill is appropriate here

Only two isolated rows are inserted over 3×33908 schedule-step records. Forward fill preserves local continuity and avoids creating artificial LR events by interpolation.

Autocorrelation Diagnosis

log-loss ACF	lag 1	lag 512	lag 2048
cosine	0.946	0.784	0.523
WSD	0.939	0.758	0.465
8-1-1	0.941	0.767	0.482

rolling residual ACF	lag 1	lag 512	lag 2048
cosine	0.994	0.176	0.148
WSD	0.999	0.852	0.640
8-1-1	1.000	0.921	0.667

Computed on steps $t \geq 1000$

Diagnosis

Adjacent loss values are not independent observations. Even after momentum correction, the smoothed residual can remain strongly autocorrelated, especially on WSD and 8-1-1 tail regions.

Effect on evaluation

- Do not use ordinary i.i.d. bootstrap over individual steps.
- Use contiguous block bootstrap with block sizes 128, 512, 2048.
- Treat bootstrap evidence as within-curve support, not schedule-level statistical significance.

Schedule Geometry Is the Stress Test

schedule	total drop	max drop	positive drops	first drop	final tail saturation
cosine	8.98×10^{-4}	8.34×10^{-8}	16453	1002	1.000
WSD	9.00×10^{-4}	6.79×10^{-7}	3391	27126	0.964
8-1-1	9.00×10^{-4}	6.84×10^{-4}	2	27126	0.964

Observation

Total LR drop is nearly matched, but decay smoothness and timing differ sharply across schedules.

Consequence

A method based only on cumulative LR mass should miss WSD tail shape and decay-local residuals.

Chapter II

Reproduced Scaling-Law Baselines

What do existing LRS-aware scaling laws explain, and what residual structure remains?

Baseline 1: Momentum Law

Tissue et al. model the loss decrease from accumulated training progress and an annealing-induced momentum term [5]:

$$\hat{L}_{\text{mom}}(t) = L_0 + AS_1(t)^{-\alpha} - CS_2(t).$$

$$S_1(t) = \sum_{i=1}^t \eta_i, \quad m_t = \rho m_{t-1} + (\eta_{t-1} - \eta_t)_+, \quad S_2(t) = \sum_{i=1}^t m_i.$$

Variables

L_0, A, C, α, ρ are fitted parameters; $S_1(t)$ is cumulative LR progress; m_t is an exponentially smoothed positive LR drop; and $S_2(t)$ accumulates the annealing signal. We use $(x)_+ = \max(x, 0)$.

Baseline 2: Multi-Power Law

Luo et al. add an explicit response to individual LR-drop events [4]:

$$\begin{aligned}\hat{L}_{\text{MPL}}(t) = & L_0 + AS_1(t)^{-\alpha} \\ & - B \sum_{k:\tau_k \leq t} \Delta\eta_k \left[1 - (1 + C\eta_{\tau_k}^{-\gamma} S_k(t))^{-\beta} \right].\end{aligned}$$

Variables

τ_k is the step of the k -th LR drop, $\Delta\eta_k = (\eta_{\tau_k-1} - \eta_{\tau_k})_+$, and $S_k(t) = \sum_{i=\tau_k}^t \eta_i$ is the post-event LR area. B, C, β, γ are fitted coefficients controlling the magnitude and saturation of each event response.

Reproduction Protocol for Baselines

Fitting objective

We fit the baseline parameters on the cosine curve by minimizing a robust log-loss discrepancy:

$$\min_{\Theta} \sum_{t \in \mathcal{T}_{\text{cos}}} H_{\delta} \left(\log L_{\text{cos}}(t) - \log \hat{L}_{\Theta}(t) \right).$$

Here H_{δ} is the Huber loss.

Transfer target

- Fit on cosine only.
- Evaluate transferred predictions on WSD.
- Report endpoint error separately because average trajectory accuracy can hide endpoint drift.

Reproduction Result 1: Momentum Law

eval schedule	MAE	R^2	endpoint abs.
cosine	0.032224	0.953201	0.025074
WSD	0.037844	0.925669	0.022260

Fit on cosine using Huber loss in log space,
evaluation uses steps $t \geq 1000$.

What it establishes

The momentum law captures a large part of the cross-schedule trend, but its WSD residual error is not random noise. This motivates modeling the remaining structure after the baseline has been fitted.

Reproduction Result 2: Multi-Power Law

WSD model	MAE	R^2	endpoint abs.
sampled momentum	0.037216	0.928180	0.045104
MPL reproduction	0.036027	0.935676	0.134094

The MPL reproduction uses sampled points and reduced training steps; it is treated as a reproduced baseline for this dataset.

What it adds

Explicit event response improves average WSD trajectory prediction slightly.

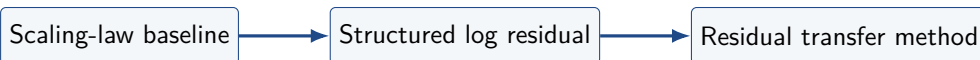
What remains

Endpoint error worsens substantially, so event response alone is insufficient for full-curve transfer.

From Baselines to Residual Modeling

Baseline laws are necessary but not sufficient

- Momentum law gives a stable physics-inspired global trend [5].
- Multi-Power Law tests the value of explicit decay-event response [4].
- Both leave enough WSD structure to justify a residual-first method.



Chapter III

Residual Method Development

Which residual coordinate transfers across schedules, and how should tail geometry enter?

Residual Target

Define the post-momentum residual:

$$r_s(t) = \log L_s(t) - \log \hat{L}_{\text{mom},s}(t).$$

Our residual model predicts

$$\log \hat{L}_s(t) = \log \hat{L}_{\text{mom},s}(t) + \hat{r}_s(t), \quad \hat{L}_s(t) = \hat{L}_{\text{mom},s}(t) \exp\{\hat{r}_s(t)\}.$$

Why model log residuals?

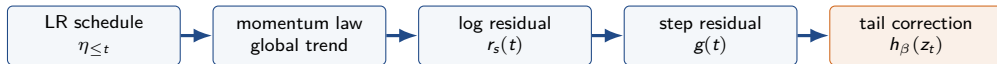
The correction is multiplicative on the original loss scale, which keeps the prediction positive and lets the baseline carry the global magnitude of the curve.

Model Selection Lessons

candidate residual model	what it tested	outcome
constant shift	scalar bias after momentum	barely improves WSD
feature ridge	hand-crafted LR summaries	weak transfer
high-dimensional local features	more LR history	overfits / worse transfer
raw or normalized S_1	intrinsic-time residual coordinate	phase-warped
MPL-style hybrid decay	low-rank saturated decay features	small gain only

These ablations motivate a low-dimensional residual model instead of direct high-dimensional schedule-feature regression.

Method Pipeline



$$\log \hat{L}(t) = \log \hat{L}_{\text{mom}}(t) + g(t) + h_\beta(z_t).$$

Modeling principle

Keep the scaling-law baseline as the interpretable global component, then add only residual terms that are selected on 8-1-1 and evaluated on held-out WSD.

Method I: Step-Aligned Residual Spline

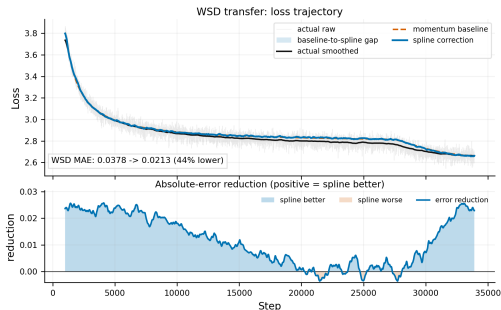
Fit a smooth function g on cosine residuals:

$$g^* = \arg \min_{g \in \mathcal{G}} \sum_{t \in \mathcal{T}_{\cos}} (r_{\cos}(t) - g(t))^2 + \lambda \int |g''(u)|^2 du.$$

Transfer:

$$\begin{aligned} \log \hat{L}_{\text{WSD}}(t) = & \log \hat{L}_{\text{mom}, \text{WSD}}(t) \\ & + \text{clip}(\gamma g^*(t), -c, c). \end{aligned}$$

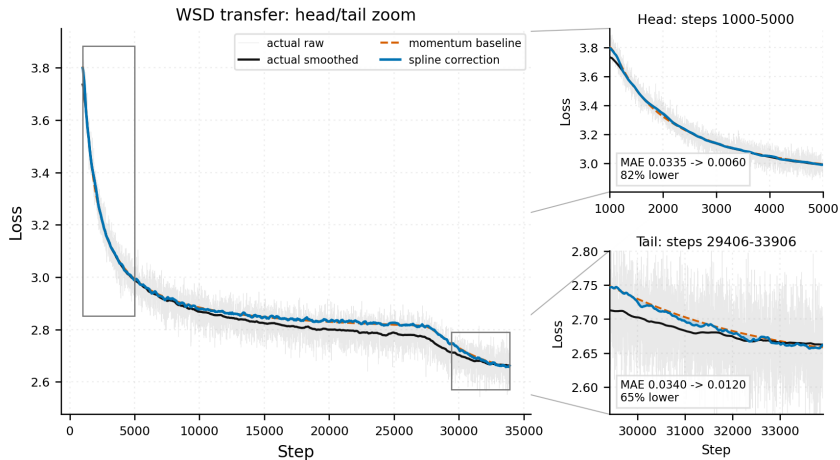
g is a cubic smoothing spline, λ controls smoothness, γ is a shrinkage coefficient, and c bounds the residual magnitude.



Top: loss trajectory with the baseline-to-spline gap shaded.

Bottom: absolute-error reduction; positive blue area means the spline correction is closer to the raw loss.

Where the Spline Helps: Head and Tail



Local zooms: the spline gain is concentrated in the early transient and late WSD decay windows. Shaded bands show the baseline-to-spline gap.

Step Spline Result and Mechanism

WSD sampled model	MAE	RMSE	R^2
momentum	0.037216	0.046672	0.928180
mean shift	0.036862	0.046243	0.929493
spline $s = 0.1$	0.020657	0.024735	0.979827

Cosine-fitted residual spline transferred to WSD.

coordinate	residual corr.	WSD MAE
absolute step	0.936	0.020767
raw S_1	-0.123	0.045414
S_1 ratio	-0.008	0.039812

After momentum correction, residual phase is step-aligned rather than S_1 -aligned.

Hypothesis Test I: Is the Spline Residual Real?

Test target

For each WSD time window W , compare paired absolute errors:

$$d_t = |L_t - \hat{L}_{\text{mom}}(t)| - |L_t - \hat{L}_{\text{spl}}(t)|, \quad \Delta_W = \frac{1}{|W|} \sum_{t \in W} d_t.$$

$$H_0 : \Delta_W \leq 0 \quad \text{vs.} \quad H_1 : \Delta_W > 0.$$

Why block bootstrap

Loss errors are strongly autocorrelated across adjacent steps, so ordinary i.i.d. bootstrap is too optimistic. We resample contiguous WSD time blocks of size 128, 512, 2048.

What would falsify it

- $q_{0.05}(\Delta_W) \leq 0$ on key windows.
- Reversed/shifted/permutated residual templates work equally well.
- 811-selected spline rank does not match WSD rank.

Spline Robustness: Bootstrap and Placebo Controls

WSD window	mean Δ	q05, block 2048	$P(\Delta > 0)$
full	0.016843	0.010497	1.000
tail 27126-33906	0.016898	0.010157	1.000
last 2048	0.024261	0.024261	1.000

Statistic: $|e_{\text{momentum}}| - |e_{\text{spline}}|$. Positive means spline improves.

residual template	full MAE	tail MAE
true step spline	0.020878	0.021855
reversed residual	0.049905	0.047814
circular shift	0.050053	0.051580
block permutation	0.050067	0.049095
sign flip	0.067432	0.067303

Placebo templates fail, so the effect is not arbitrary
residual magnitude.

Spline Robustness: Selection and Parameter Stability

config	811 MAE	WSD MAE	val rank	test rank
$s = 0.01, \lambda = 1$	0.016447	0.020878	1	1
$s = 0.05, \lambda = 1$	0.017002	0.021281	3	3
$s = 0.10, \lambda = 1$	0.017600	0.021741	5	5

Selection audit

- Config is selected by 811, not WSD.
- 811/WSD grid rank correlation:

$$\rho_{\text{Spearman}} = 0.989, \quad r_{\text{Pearson}} = 0.997.$$

- Best $s = 0.01$ and conservative $s = 0.1$ differ by less than 0.001 WSD MAE.

Conclusion: report a stable spline family, not a claim that one smoothing constant is theoretically special.

Method II: Event/Tail Leftover

Motivated by Su Jianlin's sliding-average view of weight decay and learning-rate schedules [7], we model the step template's tail leftover with a schedule-history proxy:

$$\begin{aligned}\ell_{\cos}(t) &= r_{\cos}(t) - g^*(t), & h_{\beta}(z_t) &= \beta_0 + \beta^{\top} \text{standardize}(z_t), \\ \widehat{r}_{\text{final}}(t) &= \text{clip}(g^*(t) + h_{\beta}(z_t), -c, c) .\end{aligned}$$

Feature vector z_t

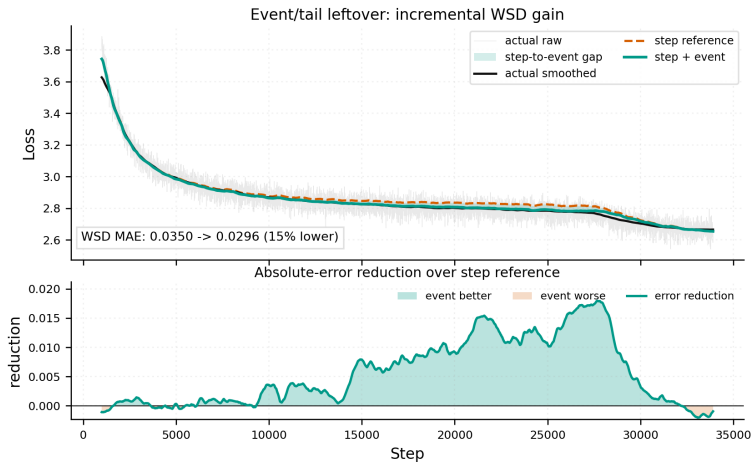
$z_t = (q_t, 1 - t/T, p_t, p_t^2)$, where p_t is normalized progress through the decay/tail region and q_t is linear decay progress. $\text{standardize}(\cdot)$ uses cosine statistics.

Interpretation boundary

z_t is not an optimizer-state measurement. It is a schedule-derived proxy for tail geometry; optimizer EMA, weight decay state, weight RMS, and update RMS are not observed in the provided data.

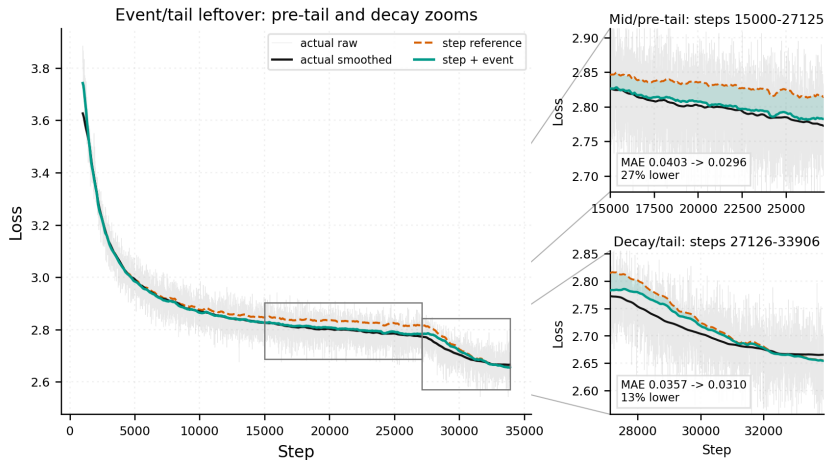
The endpoint/tail coordinates also follow the linear-decay endpoint-loss motivation in [8]; we use only LR-derived proxies because optimizer states are unobserved.

Event/Tail Leftover: Incremental Gain



The advanced correction is evaluated relative to the step reference. Positive teal area means the event/tail leftover reduces absolute prediction error beyond the spline-only model.

Where Event/Tail Helps: Mid and Tail



The gain is localized after the early transient: mid/pre-tail structure improves by 27%, and WSD decay/tail improves by 13% over the step reference.

Event/Tail Feature Evidence

Incremental information test

comparison	full MAE	interpretation
step reference	0.035007	binned reference
aligned leftover	0.029644	supported
train on 8-1-1 only	0.030152	weak transfer
train on cosine+8-1-1	0.029322	supported

The useful correction is schedule-aligned tail geometry on top of the step template.

Feature-family ablation

feature family	full MAE	tail MAE
linear endpoint	0.029644	0.031012
drop impulse	0.033357	0.037049
event-tail interactions	0.034400	0.031355
event history	0.082043	0.266411
full event-local	0.085251	0.284405

Low-dimensional endpoint/tail features transfer better than high-dimensional event-history features.

What the method development contributes

- Residual learning is anchored to a reproduced scaling-law baseline.
- The first useful residual coordinate is absolute step, not raw or normalized S_1 .
- The second useful correction is low-rank tail geometry, not arbitrary event-history features.
- The method stays testable: cosine fit, 8-1-1 selection, WSD held-out reporting.

Chapter IV

Evaluation and Limitations

How accurate is the final model on held-out WSD, and where are its limits?

Held-Out WSD Evaluation

WSD test model	full MAE	RMSE	full R^2	tail MAE	endpoint abs.
momentum baseline	0.037720	0.047275	0.926313	0.038754	0.045668
event decay only	0.032596	–	0.945083	0.032285	0.050142
step reference	0.021281	0.025610	0.978375	0.022268	0.004676
step + event leftover	0.011191	0.014137	0.993410	0.015937	0.016615

Main result

The step + event leftover model gives the best full-curve and tail accuracy on the held-out WSD schedule, with full-resolution $R^2 = 0.993410$.

Endpoint Behavior Remains a Separate Objective

Full/tail gain

$\text{MAE}_{\text{full}} : 0.021281 \rightarrow 0.011191$

$\text{MAE}_{\text{tail}} : 0.022268 \rightarrow 0.015937$

Endpoint tradeoff

endpoint abs. : $0.004676 \rightarrow 0.016615$

Pure step reference remains better for endpoint behavior.

The final claim is therefore about full-curve accuracy.

Hypothesis Test II: Does Event Geometry Add Information?

Main hypothesis

After fitting the step/spline residual, event-tail features carry additional schedule-local information:

$$H_0 : \text{MAE}(\hat{L}_{\text{step+event}}) \geq \text{MAE}(\hat{L}_{\text{step}}) \quad H_1 : \text{MAE}(\hat{L}_{\text{step+event}}) < \text{MAE}(\hat{L}_{\text{step}}).$$

test	key numbers	verdict
aligned WSD margin	full 0.035007 $\rightarrow$ 0.029644 tail 0.035694 $\rightarrow$ 0.031012	supported
train on 811 only	full 0.029869 $\rightarrow$ 0.030152	not supported
train on cosine+811	full 0.031382 $\rightarrow$ 0.029322	supported
endpoint guard	endpoint stays $\approx$ 0.03149 full 0.029716	mixed

This narrows the claim: the leftover is useful as a cosine/step-phase tail correction, not as a universal event model from any source schedule.

Event Leftover Robustness: Feature Families

feature family	WSD full MAE	WSD tail MAE
linear endpoint	0.029644	0.031012
drop impulse	0.033357	0.037049
event-tail interactions	0.034400	0.031355
event history	0.082043	0.266411
full event-local	0.085251	0.284405

Interpretation

- The gain is driven by `linear_endpoint`: decay progress, tail progress, endpoint pressure, remaining fraction.
- Direct LR-drop impulse features are weaker.
- High-dimensional event-history features overfit and transfer poorly.

No-Lookahead and Scope of Claims

Leakage checks

- WSD is not used for fitting or validation selection.
- Full candidate grids store train/validation metrics only.
- WSD test metrics are written only for selected configs.
- Transfer table has no row with WSD inside the training schedule.

What the tests prove

- Spline: robust within-curve improvement over momentum on WSD.
- Event leftover: aligned tail geometry improves full/tail MAE over step reference.
- Not proved: schedule-level statistical significance from only three schedules.
- Not claimed: endpoint-loss improvement for the event leftover.

Event Leftover: Robustness Summary

binned WSD variant	full MAE	tail MAE
step reference	0.035007	0.035694
aligned leftover	0.029644	0.031012
zero event control	0.035032	0.035836
shifted control	0.035132	0.036998
reverse-time control	0.038455	0.036350

WSD window	mean gain	q05	$P>0$
full	0.005363	0.003307	1.000
tail	0.004682	0.001656	1.000
last 512	-0.000188	-0.000487	0.000

Bootstrap evidence is within-curve support, not schedule-level significance.

Takeaway

Aligned event/tail geometry gives full and tail gains; shifted or reversed controls fail. The endpoint remains a separate objective.

Empirical Findings

Supported by experiments

- Baseline-informed residual correction.
- Absolute-step residual phase.
- Low-rank linear-tail/event leftover.
- Validation-only selection with WSD held out.

Not supported as main mechanisms

- Constant residual shift.
- Direct high-dimensional schedule features.
- Replacing step residual by S_1 -intrinsic time alone.
- Event-history features without step alignment.

Limitations and Future Work

Limitations

- Only three schedules and one model/data scale.
- Endpoint remains a separate failure mode.
- LR schedule is only a proxy for unobserved optimizer state.
- Some reproduction settings are simplified under the available compute budget.

Next experiments

- Add explicit endpoint constraints.
- Test more model sizes, horizons, and schedules.
- Use optimizer-state traces if available.
- Replace spline with a constrained dynamics-derived basis.

A strong loss-curve predictor should use scaling laws as a base layer, then model only the residual structure that transfers.

1. Reproduce

Momentum law and Multi-Power Law provide LRS-aware baselines [4, 5].

2. Improve

Step-aligned residual spline substantially reduces WSD transfer error.

3. Refine

Event leftover reaches WSD full $R^2 = 0.993410$, with endpoint caveat.

Repository

`https://github.com/zyyyyyyy123/tdlt`

Division of Labor

member	contribution
占扬 / 2300010819 / Undergraduate	Problem framing, literature reading, Tissue et al. momentum-law reproduction, presentation background.
唐锦琪 / 2300010628 / Undergraduate	Luo et al. Multi-Power Law reproduction, data processing, baseline comparison, metric validation.
邓子钧 / 2200015833 / Undergraduate	Residual spline, event-leftover method development, validation protocol design, slide integration.

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